

FOR IBPS SBI PO/CLERK PRELIMS 2026

QUANT CHECKLIST

the PRACTICE
PAPER

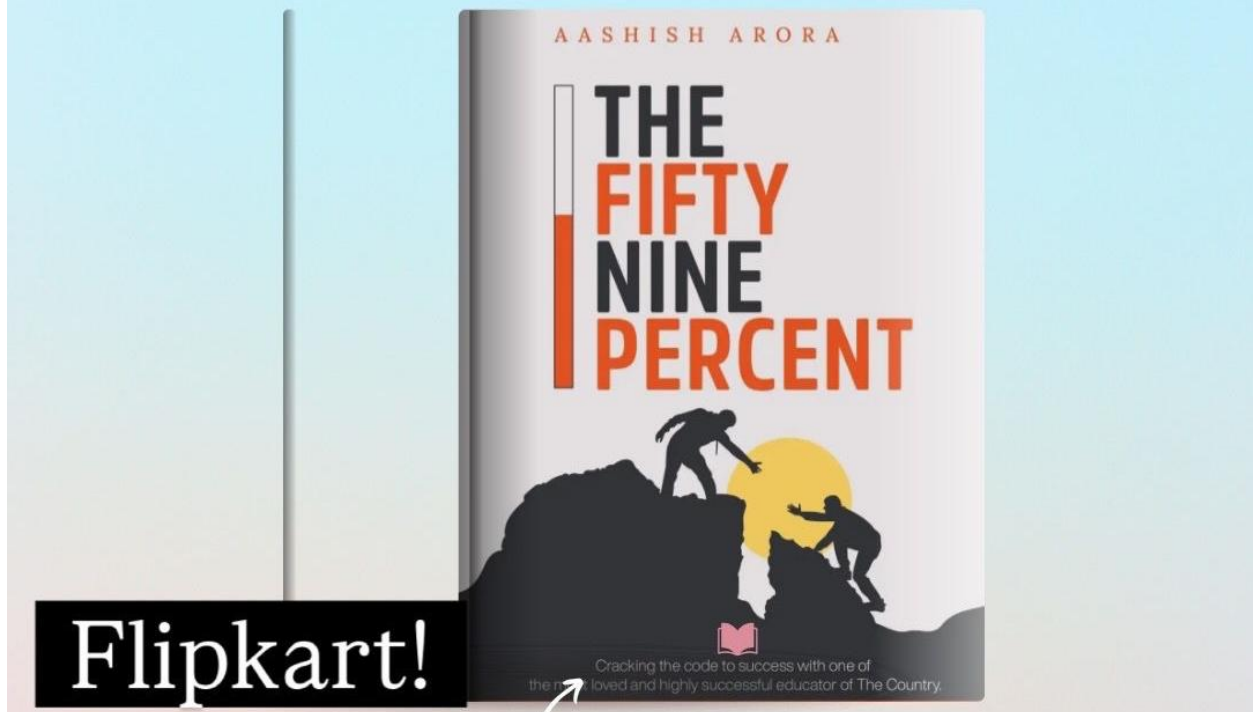


/STUDIFIEDPDF

Subscribe to
STUDIFIEDTM

**THE BOOK YOU
LOVED IS BACK**

Only at Rs.199/-



Dear Students,

This daily practice sheet is designed to build both speed and accuracy, one day at a time.

It contains a mix of easy, moderate, and challenging questions to prepare you for every possible scenario in the exam. Treat it like a warm-up before the real game.

Solve it daily without fail. Don't wait for motivation—show up with discipline. Because it's not talent but consistent hard work that takes you places.

Stay focused. Stay consistent. Work hard.

-Aashish Arora

yes
OFFICER

ALL COURSES UNDER ONE SUBSCRIPTION'

AASHISH ARORA EXCLUSIVE

MAINS COURSES

- MAINS FIRST
- THE MATHS HERO PRO
- MAINS SETU
- MAINS D.I SMARTBOOK
- LEVEL UP 1.0, 2.0
- MAINS MADE EASY 1.0, 2.0
- MAINS SARTHI
- OPERATION MAINS

FOUNDATION COURSES

- THE MATHS HERO 2026 (NEW)
- THE MATHS HERO 2025
- THE MATHS HERO 2024
- UDAAN D.I
- GANIT 1.0, 2.0
- LAKSHYA COURSE

USE CODE :- TT10

FOR MAXIMUM DISCOUNT



CONTENTS

1. SIMPLIFICATION & APPROXIMATION	9
2. ARITHMETIC WORD PROBLEMS	22
3. QUADRATIC EQUATIONS	43
4. WRONG NUMBER SERIES	59
5. MISSING NUMBER SERIES	70
6. DATA INTERPRETATION	82

1. SIMPLIFICATION AND APPROXIMATION

Direction: What value should come in place of the question mark (?) in the following question?

1. 62.5% of $896 + \sqrt{7056} - (21 \times 18) = ? \times 4 + 31$

- A. 58.75
- B. 56.75
- C. 60.75
- D. 62.75
- E. None of these

2. $?%$ of $750 + \frac{7}{12}$ of $936 = 44.44\%$ of $1215 + 18 \times 32$

- A. 72
- B. 76
- C. 80
- D. 84
- E. None of these

3. $3\frac{1}{4}$ of $160 - 37.5\%$ of $384 + \sqrt[3]{46656} = ? + 275$

- A. 129
- B. 133

C. 137

D. 141

E. None of these

4. $(\sqrt{4096} \times \sqrt{361}) \div 16 + 27.27\%$ of $1331 = 2 \times ? + 43$

- A. 188
- B. 194
- C. 204
- D. 198
- E. None of these

5. $18^2 + 56.25\%$ of $768 - \frac{5}{14}$ of $1008 = ? \times 7 + 25$

- A. 49
- B. 53
- C. 51
- D. 55
- E. None of these

6. 91.66% of $360 + \frac{1}{13}$ of $1183 + 4^3 = ? + 392$

A. 91

B. 97

C. 87

D. 93

E. None of these

7. $(41.66\% \text{ of } 1008 + 68.75\% \text{ of } 512) \div 4 + ? = 23^2$

A. 328

B. 336

C. 332

D. 340

E. None of these

8. $(17 \times 43) - (19 \times 26) + 87.5\% \text{ of } 288 = ?^2 + 48$

A. 18

B. 21

C. 19

D. 23

E. None of these

9. $\sqrt{?} + 41.66\% \text{ of } 576 = 18^2$

A. 7056

B. 6889

C. 7225

D. 7396

E. None of these

10. $2\frac{2}{5} \text{ of } 375 - 75\% \text{ of } 640 + \sqrt{2025} = ? \times 5 + 65$

A. 72

B. 76

C. 84

D. 80

E. None of these

11. $16.66\% \text{ of } 1944 + 58.33\% \text{ of } 864 - 12.5\% \text{ of } 1216 = ? + 498$

A. 168

B. 178

C. 174

D. 184

E. None of these

12. $4^3 \times 12 - 31.25\% \text{ of } 1344 + \frac{1}{9} \text{ of } 1458 = ? \times 3$

A. 170

B. 162

C. 166

D. 174

E. None of these

13. $(875 + 625) \div 15 + (35^2 - 29^2) \div 16 = ?$

A. 118

B. 120

C. 128

D. 124

E. None of these

14. $? \div 8 + 14.28\% \text{ of } 686 = 83.33\% \text{ of } 540 + \sqrt[3]{21952}$

A. 2960

B. 3120

C. 3040

D. 3200

E. None of these

15. $\sqrt{1764} + 22.22\% \text{ of } 945 + 2^3 \times 25 = ? + 310$

A. 138

B. 146

C. 134

D. 142

E. None of these

16. $25\% \text{ of } (1296 + 704) - 6.25\% \text{ of } 1600 + \sqrt[3]{64000} = ? \times 10$

A. 42

B. 44

C. 40

D. 46

E. None of these

17. $72.72\% \text{ of } 1815 - 37.5\% \text{ of } 1152 + \frac{7}{8} \text{ of } 640 = ? + 728$

A. 700

B. 680

C. 740

D. 720

E. None of these

18. $5\frac{1}{2} \text{ of } 84 + 4\frac{2}{3} \text{ of } 126 - 2\frac{3}{7} \text{ of } 147 = ? \times 6$

A. 110.5

B. 115.5

C. 120.5

D. 105.5

E. None of these

19. $56.25\% \text{ of } 448 + (35^2 - 29^2) \div 12 + 44.44\% \text{ of } 405 = ? + 210$

A. 246

B. 258

C. 254

D. 250

E. None of these

20. $24 \times \sqrt{2401} - 17 \times \sqrt[3]{19683} + 28\% \text{ of } 1250 = ? \times 11$

A. 101

B. 91

- C. 97
D. 94
E. None of these

Answers:

1. A
2. B
3. C
4. D
5. B
6. D
7. B
8. B
9. A
10. D
11. B
12. A
13. D
14. C
15. D
16. B
17. D
18. B
19. C
20. C

Solutions:

$$1. 62.5\% \text{ of } 896 + \sqrt{7056} - (21 \times 18) = ? \times 4 + 31$$

$$62.5\% = \frac{5}{8} \text{ and } \sqrt{7056} = 84$$

$$\text{So, } 560 + 84 - 378 = ? \times 4 + 31$$

$$266 = ? \times 4 + 31$$

$$? \times 4 = 235$$

$$? = 58.75$$

$$2. ?\% \text{ of } 750 + \frac{7}{12} \text{ of } 936 = 44.44\% \text{ of } 1215 + 18 \times 32$$

$$44.44\% = \frac{4}{9} \text{ and } \frac{7}{12} \text{ of } 936 = 546$$

$$\text{So, } ?\% \text{ of } 750 + 546 = 540 + 576$$

$$?\% \text{ of } 750 + 546 = 1116$$

$$?\% \text{ of } 750 = 570$$

$$? = 570 \times 100 \div 750 = 76$$

$$3. 3\frac{1}{4} \text{ of } 160 - 37.5\% \text{ of } 384 + \sqrt[3]{46656} = ? + 275$$

$$3\frac{1}{4} = \frac{13}{4}, 37.5\% = \frac{3}{8} \text{ and } \sqrt[3]{46656} = 36$$

$$\text{So, } \frac{13}{4} \text{ of } 160 - \frac{3}{8} \text{ of } 384 + 36 = ? + 275$$

$$520 - 144 + 36 = ? + 275$$

$$412 = ? + 275$$

$$? = 137$$

$$4. (\sqrt{4096} \times \sqrt{361}) \div 16 + 27.27\% \text{ of } 1331 = 2 \times ? + 43$$

$$\sqrt{4096} = 64, \sqrt{361} = 19 \text{ and } 27.27\% = \frac{3}{11}$$

$$\text{So, } (64 \times 19) \div 16 + \frac{3}{11} \text{ of } 1331 = 2 \times ? + 43$$

$$76 + 363 = 2 \times ? + 43$$

$$439 = 2 \times ? + 43$$

$$2 \times ? = 396$$

$$? = 198$$

$$5. 18^2 + 56.25\% \text{ of } 768 - \frac{5}{14} \text{ of } 1008 = ? \times 7 + 25$$

$$56.25\% = \frac{9}{16} \text{ and } 18^2 = 324$$

$$56.25\% \text{ of } 768 = 432 \text{ and } \frac{5}{14} \text{ of } 1008 = 360$$

$$\text{So, } 324 + 432 - 360 = ? \times 7 + 25$$

$$396 = ? \times 7 + 25$$

$$? \times 7 = 371$$

$$? = 53$$

$$6. 91.66\% \text{ of } 360 + \frac{1}{13} \text{ of } 1183 + 4^3 = ? + 392$$

$$91.66\% = \frac{11}{12}, \frac{1}{13} \text{ of } 1183 = 91 \text{ and } 4^3 = 64$$

$$\text{So, } 330 + 91 + 64 = ? + 392$$

$$485 = ? + 392$$

$$? = 93$$

$$7. (41.66\% \text{ of } 1008 + 68.75\% \text{ of } 512) \div 4 + ? = 23^2$$

$$41.66\% = \frac{5}{12} \text{ and } 68.75\% = \frac{11}{16}$$

$$41.66\% \text{ of } 1008 = 420 \text{ and } 68.75\% \text{ of } 512 = 352$$

$$\text{So, } (420 + 352) \div 4 + ? = 23^2$$

$$772 \div 4 + ? = 529$$

$$193 + ? = 529$$

$$? = 336$$

$$8. (17 \times 43) - (19 \times 26) + 87.5\% \text{ of } 288 = ?^2 + 48$$

$$87.5\% = \frac{7}{8} \text{ and } 17 \times 43 = 731$$

$$19 \times 26 = 494 \text{ and } 87.5\% \text{ of } 288 = 252$$

$$\text{So, } 731 - 494 + 252 = ?^2 + 48$$

$$489 = ?^2 + 48$$

$$?^2 = 441$$

$$? = 21$$

$$9. \sqrt{?} + 41.66\% \text{ of } 576 = 18^2$$

$$41.66\% = \frac{5}{12} \text{ and } 18^2 = 324$$

$$41.66\% \text{ of } 576 = 240$$

$$\text{So, } \sqrt{?} + 240 = 324$$

$$\sqrt{?} = 84$$

$$? = 7056$$

$$10. 2\frac{2}{5} \text{ of } 375 - 75\% \text{ of } 640 + \sqrt{2025} = ? \times 5 + 65$$

$$2\frac{2}{5} = \frac{12}{5}, 75\% = \frac{3}{4} \text{ and } \sqrt{2025} = 45$$

$$\text{So, } \frac{12}{5} \text{ of } 375 - \frac{3}{4} \text{ of } 640 + 45 = ? \times 5 + 65$$

$$900 - 480 + 45 = ? \times 5 + 65$$

$$465 = ? \times 5 + 65$$

$$? \times 5 = 400$$

$$? = 80$$

$$11. 16.66\% \text{ of } 1944 + 58.33\% \text{ of } 864 - 12.5\% \text{ of } 1216 = ? + 498$$

$$16.66\% = \frac{1}{6}, 58.33\% = \frac{7}{12} \text{ and } 12.5\% = \frac{1}{8}$$

$$\text{So, } \frac{1}{6} \text{ of } 1944 + \frac{7}{12} \text{ of } 864 - \frac{1}{8} \text{ of } 1216 = ? + 498$$

$$324 + 504 - 152 = ? + 498$$

$$676 = ? + 498$$

$$? = 178$$

$$12. 4^3 \times 12 - 31.25\% \text{ of } 1344 + \frac{1}{9} \text{ of } 1458 = ? \times 3$$

$$4^3 = 64 \text{ and } 31.25\% = \frac{5}{16}$$

$$31.25\% \text{ of } 1344 = 420 \text{ and } \frac{1}{9} \text{ of } 1458 = 162$$

$$\text{So, } 64 \times 12 - 420 + 162 = ? \times 3$$

$$768 - 420 + 162 = ? \times 3$$

$$510 = ? \times 3$$

$$? = 170$$

$$13. (875 + 625) \div 15 + (35^2 - 29^2) \div 16 = ?$$

$$(875 + 625) = 1500 \text{ and } (35^2 - 29^2) = 384$$

$$\text{So, } 1500 \div 15 + 384 \div 16 = ?$$

$$100 + 24 = ?$$

$$? = 124$$

$$14. ? \div 8 + 14.28\% \text{ of } 686 = 83.33\% \text{ of } 540 + \sqrt[3]{21952}$$

$$14.28\% = \frac{1}{7}, 83.33\% = \frac{5}{6} \text{ and } \sqrt[3]{21952} = 28$$

$$\text{So, } ? \div 8 + 98 = 450 + 28$$

$$? \div 8 + 98 = 478$$

$$? \div 8 = 380$$

$$? = 3040$$

$$15. \sqrt{1764} + 22.22\% \text{ of } 945 + 2^3 \times 25 = ? + 310$$

$$\sqrt{1764} = 42, 22.22\% = \frac{2}{9} \text{ and } 2^3 \times 25 = 200$$

$$22.22\% \text{ of } 945 = 210$$

$$\text{So, } 42 + 210 + 200 = ? + 310$$

$$452 = ? + 310$$

$$? = 142$$

$$16. 25\% \text{ of } (1296 + 704) - 6.25\% \text{ of } 1600 + \sqrt[3]{64000} = ? \times 10$$

$$25\% = \frac{1}{4}, 6.25\% = \frac{1}{16} \text{ and } \sqrt[3]{64000} = 40$$

$$(1296 + 704) = 2000$$

$$\text{So, } \frac{1}{4} \text{ of } 2000 - \frac{1}{16} \text{ of } 1600 + 40 = ? \times 10$$

$$500 - 100 + 40 = ? \times 10$$

$$440 = ? \times 10$$

$$? = 44$$

$$17. 72.72\% \text{ of } 1815 - 37.5\% \text{ of } 1152 + \frac{7}{8} \text{ of } 640 = ? + 728$$

$$72.72\% = \frac{8}{11}, 37.5\% = \frac{3}{8} \text{ and } \frac{7}{8} \text{ of } 640 = 560$$

$$72.72\% \text{ of } 1815 = 1320 \text{ and } 37.5\% \text{ of } 1152 = 432$$

$$\text{So, } 1320 - 432 + 560 = ? + 728$$

$$1448 = ? + 728$$

$$? = 720$$

$$18. 5\frac{1}{2} \text{ of } 84 + 4\frac{2}{3} \text{ of } 126 - 2\frac{3}{7} \text{ of } 147 = ? \times 6$$

$$5\frac{1}{2} = \frac{11}{2}, 4\frac{2}{3} = \frac{14}{3} \text{ and } 2\frac{3}{7} = \frac{17}{7}$$

$$\text{So, } \frac{11}{2} \text{ of } 84 + \frac{14}{3} \text{ of } 126 - \frac{17}{7} \text{ of } 147 = ? \times 6$$

$$462 + 588 - 357 = ? \times 6$$

$$693 = ? \times 6$$

$$? = 115.5$$

$$19. 56.25\% \text{ of } 448 + (35^2 - 29^2) \div 12 + 44.44\% \text{ of } 405 = ? + 210$$

$$56.25\% = \frac{9}{16} \text{ and } 44.44\% = \frac{4}{9}$$

$$(35^2 - 29^2) = 384$$

$$\text{So, } \frac{9}{16} \text{ of } 448 + 384 \div 12 + \frac{4}{9} \text{ of } 405 = ? + 210$$

$$252 + 32 + 180 = ? + 210$$

$$464 = ? + 210$$

$$? = 254$$

$$20. 24 \times \sqrt{2401} - 17 \times \sqrt[3]{19683} + 28\% \text{ of } 1250 = ? \times 11$$

$$\sqrt{2401} = 49, \sqrt[3]{19683} = 27 \text{ and } 28\% \text{ of } 1250 = 350$$

$$\text{So, } 24 \times 49 - 17 \times 27 + 350 = ? \times 11$$

$$1176 - 459 + 350 = ? \times 11$$

$$1067 = ? \times 11$$

$$? = 97$$

2. ARITHMETIC QUESTIONS

Q1: Monthly income of a person is Rs 50,000 and his monthly expenditure is Rs 30,000. If his income increases by 24% and his savings increase by 30%, then find the percentage increase in his monthly expenditure.

एक व्यक्ति की मासिक आय Rs 50,000 है और उसका मासिक खर्च Rs 30,000 है। यदि उसकी आय 24% बढ़ जाती है और उसकी बचत 30% बढ़ जाती है, तो उसके मासिक खर्च में प्रतिशत वृद्धि ज्ञात कीजिए।

- (A) 16%
- (B) 18%
- (C) 20%
- (D) 22%
- (E) 25%

Q2: Kartik bought a second-hand camera and spent Rs 2,000 on its repair. He sold it for Rs 24,000 and earned a profit of 20%. If he had not repaired it and sold it at a profit of 15%, then find the new selling price of the camera.

कार्तिक ने एक सेकंड-हैंड कैमरा खरीदा और उसकी मरम्मत पर Rs 2,000 खर्च किए। उसने उसे Rs 24,000 में बेचकर 20% लाभ कमाया। यदि उसने उसकी मरम्मत नहीं करवाई होती और उसे 15% लाभ पर बेचा होता, तो कैमरे का नया विक्रय मूल्य ज्ञात कीजिए।

- (A) Rs 20,400
- (B) Rs 19,800
- (C) Rs 21,600
- (D) Rs 20,250
- (E) Rs 20,700

Q3: Four years hence, the ratio of ages of Aniket and Divya will be equal to the ratio of ages of Kunal and Divya two years ago. If Aniket is 6 years older than Kunal and Divya's present age is 20 years, then find Kunal's present age.

चार वर्ष बाद अनिकेत और दिव्या की आयु का अनुपात, दो वर्ष पहले कुणाल और दिव्या की आयु के अनुपात के बराबर होगा। यदि अनिकेत, कुणाल से 6 वर्ष बड़ा है और दिव्या की वर्तमान आयु 20 वर्ष है, तो कुणाल की वर्तमान आयु ज्ञात कीजिए।

- (A) 34 years
- (B) 38 years

- (C) 42 years
- (D) 36 years
- (E) 40 years

Q4: In mixture A, the quantity of milk is 30 liters, which is 150% of the quantity of water. In mixture B, the quantity of milk is 28 liters, which is 70% of the quantity of water. If both mixtures are mixed to form mixture C, then find the difference between the quantities of milk and water in the resulting mixture.

मिश्रण A में दूध की मात्रा 30 लीटर है, जो पानी की मात्रा का 150% है। मिश्रण B में दूध की मात्रा 28 लीटर है, जो पानी की मात्रा का 70% है। यदि दोनों मिश्रणों को मिलाकर मिश्रण C बनाया जाता है, तो परिणामी मिश्रण में दूध और पानी की मात्राओं के बीच अंतर ज्ञात कीजिए।

- (A) 6 liters
- (B) 4 liters
- (C) 8 liters
- (D) 2 liters
- (E) 10 liters

Q5: A boat covers 160 km downstream in 8 hours. The ratio of the speed of the boat in still water to the speed of the stream is 4:1. If the speed of the stream is increased by 50%, then find the time taken by the boat to cover 90 km upstream.

एक नाव धारा के अनुकूल 160 किमी की दूरी 8 घंटों में तय करती है। शांत जल में नाव की गति और धारा की गति का अनुपात 4:1 है। यदि धारा की गति 50% बढ़ा दी जाती है, तो नाव द्वारा धारा के प्रतिकूल 90 किमी की दूरी तय करने में लिया गया समय ज्ञात कीजिए।

- (A) 9 hours
- (B) 10 hours
- (C) 12 hours
- (D) 8 hours
- (E) 15 hours

Q6: A retailer marks a bag 75% above its cost price and allows a discount of 20%. If he still earns a profit of Rs. 800, find the selling price of the bag.

एक दुकानदार एक बैग का अंकित मूल्य उसके क्रय मूल्य से 75% अधिक रखता है और 20% की छूट देता है। यदि उसे फिर भी Rs. 800 का लाभ होता है, तो बैग का विक्रय मूल्य ज्ञात कीजिए।

- (A) Rs. 2400
- (B) Rs. 3200
- (C) Rs. 3000

- (D) Rs. 2800
- (E) Rs. 2600

Q7: In an election between two candidates, Lokesh gets 25% fewer votes than Sandeep. If Sandeep wins by 900 votes, find the total number of votes polled.
दो उम्मीदवारों के बीच हुए चुनाव में लोकेश को संदीप से 25% कम वोट मिलते हैं। यदि संदीप 900 वोटों से जीतता है, तो कुल डाले गए वोटों की संख्या ज्ञात कीजिए।

- (A) 6300
- (B) 7200
- (C) 5400
- (D) 6000
- (E) 8100

Q8: Pipes M and N together can fill $\frac{1}{3}$ of a tank in 5 hours, while Pipe S can empty the full tank in 30 hours. If Pipes M and N are opened together for 10 hours, find the time taken by Pipe S to empty the filled part of the tank.

पाइप M और N मिलकर एक टैंक का $\frac{1}{3}$ भाग 5 घंटे में भर सकते हैं, जबकि पाइप S पूरे भरे हुए टैंक को 30 घंटे में खाली कर सकता है। यदि पाइप M और N को 10 घंटे के लिए साथ खोला जाए, तो पाइप S को भरे हुए भाग को खाली करने में कितना समय लगेगा?

- (A) 15 hours
- (B) 18 hours
- (C) 24 hours
- (D) 30 hours
- (E) 20 hours

Q9: The circumference of a circle with radius 14 cm is 16 cm less than the perimeter of a square. Find the area of the square. Use $\pi = \frac{22}{7}$.

14 सेमी त्रिज्या वाले एक वृत्त की परिधि, एक वर्ग के परिमाप से 16 सेमी कम है। वर्ग का क्षेत्रफल ज्ञात कीजिए। $\pi = \frac{22}{7}$ का प्रयोग कीजिए।

- (A) 625 sq. cm
- (B) 576 sq. cm
- (C) 676 sq. cm
- (D) 729 sq. cm
- (E) 784 sq. cm

Q10: The ratio of the speed of Train X to that of Train Y is 5 : 4. The lengths of Train X and Train Y are 900 meters and 1260 meters respectively. If both trains

are running in opposite directions and cross each other in 24 seconds, find the speed of Train X.

ट्रेन X और ट्रेन Y की गति का अनुपात 5 : 4 है। ट्रेन X और ट्रेन Y की लंबाई क्रमशः 900 मीटर और 1260 मीटर है। यदि दोनों ट्रेनें विपरीत दिशाओं में चलती हुई 24 सेकंड में एक-दूसरे को पार करती हैं, तो ट्रेन X की गति ज्ञात कीजिए।

- (A) 40 m/sec
- (B) 50 m/sec
- (C) 60 m/sec
- (D) 45 m/sec
- (E) 55 m/sec

Q11: Two partners M and N started a business by investing in the ratio of 5 : 8 respectively. M received Rs. 6000 as profit out of the total profit of Rs. 14000. If M invested for 12 months, find the number of months for which N invested.

दो साझेदार M और N ने क्रमशः 5 : 8 के अनुपात में निवेश करके एक व्यवसाय शुरू किया। कुल Rs. 14000 के लाभ में से M को Rs. 6000 लाभांश प्राप्त हुआ। यदि M ने 12 महीनों के लिए निवेश किया, तो N ने कितने महीनों के लिए निवेश किया?

- (A) 8 months
- (B) 9 months
- (C) 10 months
- (D) 12 months
- (E) 15 months

Q12: A dealer marked an item 75% above its cost price and allowed a discount of 30% on it. Had he allowed only 18% discount, he would have earned Rs. 210 more. Find the cost price of the item.

एक व्यापारी ने एक वस्तु का अंकित मूल्य उसके क्रय मूल्य से 75% अधिक रखा और उस पर 30% की छूट दी। यदि उसने केवल 18% की छूट दी होती, तो उसे Rs. 210 अधिक प्राप्त होते। वस्तु का क्रय मूल्य ज्ञात कीजिए।

- (A) Rs. 900
- (B) Rs. 1000
- (C) Rs. 1200
- (D) Rs. 800
- (E) Rs. 1100

Q13: Two persons invested equal amounts. The first person invested the whole amount at 20% per annum compound interest for 2 years. The second person

invested half of his amount at $R\%$ per annum simple interest for 2 years and the remaining half at 10% per annum simple interest for 2 years. If the total interest received by both persons is equal, find R .

दो व्यक्तियों ने समान राशि निवेश की। पहले व्यक्ति ने पूरी राशि 2 वर्षों के लिए 20% वार्षिक चक्रवृद्धि ब्याज पर निवेश की। दूसरे व्यक्ति ने अपनी आधी राशि 2 वर्षों के लिए $R\%$ वार्षिक साधारण ब्याज पर और शेष आधी राशि 2 वर्षों के लिए 10% वार्षिक साधारण ब्याज पर निवेश की। यदि दोनों व्यक्तियों को प्राप्त कुल ब्याज समान है, तो R ज्ञात कीजिए।

- (A) 30
- (B) 28
- (C) 34
- (D) 36
- (E) 32

Q14: Pipe P can fill half of a tank in 6 minutes. Pipe Q is 100% more efficient than Pipe P, and Pipe R is 50% less efficient than Pipe Q. Find the time taken by all three pipes together to fill the tank.

पाइप P एक टंकी का आधा भाग 6 मिनट में भर सकता है। पाइप Q, पाइप P से 100% अधिक कुशल है और पाइप R, पाइप Q से 50% कम कुशल है। तीनों पाइप मिलकर टंकी को भरने में कितना समय लेंगे?

- (A) 2 minutes
- (B) 3 minutes
- (C) 4 minutes
- (D) 5 minutes
- (E) 6 minutes

Q15: Vessel P contains a mixture of milk and water in which milk is 40% of the total mixture. Vessel Q contains a mixture of milk and water in which the quantity of milk is 36 liters. If the ratio of the quantity of mixture in Vessel P to that in Vessel Q is $3 : 5$, and the final mixture contains 30% milk after mixing both vessels, find the total quantity of the final mixture.

बर्तन P में दूध और पानी का मिश्रण है, जिसमें दूध कुल मिश्रण का 40% है। बर्तन Q में दूध और पानी का मिश्रण है, जिसमें दूध की मात्रा 36 लीटर है। यदि बर्तन P और बर्तन Q में मिश्रण की मात्रा का अनुपात $3 : 5$ है और दोनों मिश्रणों को मिलाने के बाद अंतिम मिश्रण में 30% दूध है, तो अंतिम मिश्रण की कुल मात्रा ज्ञात कीजिए।

- (A) 180 liters
- (B) 200 liters
- (C) 240 liters

- (D) 300 liters
- (E) 320 liters

Q16: A tank contains 150 liters of a mixture of acid and water in the ratio of 2:3, respectively. When X liters of the mixture is taken out and replaced with $(X - 10)$ liters of water, the ratio of acid to water becomes 5:9. Find X.

एक टैंक में अम्ल और पानी का 150 लीटर मिश्रण क्रमशः 2:3 के अनुपात में है। जब मिश्रण में से x लीटर निकालकर उसकी जगह $(x - 10)$ लीटर पानी डाला जाता है, तो अम्ल और पानी का अनुपात 5:9 हो जाता है। x ज्ञात कीजिए।

- (A) 15
- (B) 20
- (C) 25
- (D) 30
- (E) 35

Q17: Kunal and Tarun started a business by investing their capitals in the ratio of 4:7, respectively. After six months, Kunal increased his initial investment by 50%, while Tarun withdrew $\frac{3}{7}$ of his initial investment. If the difference between their profit shares at the end of the year is Rs. 1200, then find Tarun's profit share.

कुणाल और तरुण ने क्रमशः 4:7 के अनुपात में पूंजी लगाकर एक व्यवसाय शुरू किया। छह महीने बाद, कुणाल ने अपने प्रारंभिक निवेश में 50% की वृद्धि की, जबकि तरुण ने अपने प्रारंभिक निवेश का $\frac{3}{7}$ भाग निकाल लिया। यदि वर्ष के अंत में उनके लाभांशों के बीच का अंतर Rs. 1200 है, तो तरुण का लाभांश ज्ञात कीजिए।

- (A) Rs. 10800
- (B) Rs. 12000
- (C) Rs. 14400
- (D) Rs. 13200
- (E) Rs. 15600

Q18: Three years ago, the average age of Neha, Ritu and Varun was 28 years. If four years hence, the sum of the ages of Neha and Varun will be 68 years, then find the present age of Ritu.

तीन वर्ष पहले, नेहा, रितु और वरुण की औसत आयु 28 वर्ष थी। यदि चार वर्ष बाद नेहा और वरुण की आयु का योग 68 वर्ष होगा, तो रितु की वर्तमान आयु ज्ञात कीजिए।

- (A) 30 years
- (B) 31 years
- (C) 33 years

- (D) 36 years
(E) 39 years

Q19: Meera spends 30% of her monthly salary on rent and 20% of the remaining salary on groceries. She saves the rest of the salary. If her monthly saving is Rs. 11200, then find her monthly salary.

मीरा अपनी मासिक आय का 30% किराए पर और बची हुई आय का 20% किराने पर खर्च करती है। वह शेष आय बचाती है। यदि उसकी मासिक बचत Rs. 11200 है, तो उसकी मासिक आय ज्ञात कीजिए।

- (A) Rs. 18000
(B) Rs. 20000
(C) Rs. 22000
(D) Rs. 24000
(E) Rs. 25000

Q20: The ratio of the downstream speed of a boat to its upstream speed is 5:1. The total time taken by the boat to cover D km downstream and D km upstream together is 18 hours. Find the time taken by the boat to cover 3D km in still water.
एक नाव की धारा के अनुकूल गति और धारा के प्रतिकूल गति का अनुपात 5:1 है। नाव द्वारा D किमी धारा के अनुकूल और D किमी धारा के प्रतिकूल तय करने में कुल 18 घंटे लगते हैं। शांत जल में 3D किमी दूरी तय करने में नाव को कितना समय लगेगा?

- (A) 12 hours
(B) 15 hours
(C) 18 hours
(D) 20 hours
(E) 24 hours

Answers:

- | | |
|------|------|
| 1. C | 10.B |
| 2. E | 11.C |
| 3. B | 12.B |
| 4. D | 13.C |
| 5. A | 14.B |
| 6. D | 15.C |
| 7. A | 16.C |
| 8. E | 17.D |
| 9. C | 18.C |

19.B

20.B

Solutions:**1. Option C**

Initial monthly income = Rs 50,000.

Initial monthly expenditure = Rs 30,000.

Initial monthly savings = Income – Expenditure

Initial savings = $50,000 - 30,000 = \text{Rs } 20,000$.

Income increases by 24%.

New income = $50,000 \times 124/100$

New income = Rs 62,000.

Savings increase by 30%.

New savings = $20,000 \times 130/100$

New savings = Rs 26,000.

New expenditure = New income – New savings

New expenditure = $62,000 - 26,000$

New expenditure = Rs 36,000.

Increase in expenditure = $36,000 - 30,000$

Increase in expenditure = Rs 6,000.

Percentage increase in expenditure = $6,000 \times 100/30,000$

Percentage increase = 20%.

2. Option E

Selling price after repair = Rs 24,000.

Profit = 20%.

So, selling price = 120% of total cost price.

Total cost price = $24,000 \times 100/120$

Total cost price = Rs 20,000.

Repair cost = Rs 2,000.

Original cost price of camera = $20,000 - 2,000$

Original cost price = Rs 18,000.

If the camera was not repaired, cost price = Rs 18,000.

Profit in the new case = 15%.

New selling price = $18,000 \times 115/100$

New selling price = Rs 20,700.

3. Option B

Let Kunal's present age be x years.

Aniket is 6 years older than Kunal.

So, Aniket's present age = $x + 6$ years.

Divya's present age = 20 years.

Four years hence:

Aniket's age = $x + 6 + 4 = x + 10$ years.

Divya's age = $20 + 4 = 24$ years.

Two years ago:

Kunal's age = $x - 2$ years.

Divya's age = $20 - 2 = 18$ years.

According to the question:

$$(x + 10)/24 = (x - 2)/18$$

Cross multiply:

$$18 \times (x + 10) = 24 \times (x - 2)$$

$$18x + 180 = 24x - 48$$

$$180 + 48 = 24x - 18x$$

$$228 = 6x$$

$$x = 228/6$$

$$x = 38.$$

4. Option D

In mixture A, milk = 30 liters.

Milk is 150% of water.

$$\text{So, } 30 = 150/100 \times \text{water}$$

$$30 = 3/2 \times \text{water}$$

$$\text{Water} = 30 \times 2/3$$

$$\text{Water} = 20 \text{ liters.}$$

In mixture B, milk = 28 liters.

Milk is 70% of water.

$$\text{So, } 28 = 70/100 \times \text{water}$$

$$28 = 7/10 \times \text{water}$$

$$\text{Water} = 28 \times 10/7$$

$$\text{Water} = 40 \text{ liters.}$$

Now both mixtures are mixed.

Total milk = $30 + 28 = 58$ liters.
 Total water = $20 + 40 = 60$ liters.
 Required difference = $60 - 58$
 Required difference = 2 liters.

5. Option A

Downstream distance = 160 km.
 Downstream time = 8 hours.
 Downstream speed = $160/8$
 Downstream speed = 20 km/hr.
 Ratio of speed of boat in still water to speed of stream = 4:1.
 Let speed of boat in still water = $4x$.
 Let speed of stream = x .
 Downstream speed = $4x + x = 5x$.
 So, $5x = 20$.
 $x = 20/5$
 $x = 4$.
 Speed of stream = 4 km/hr.
 Speed of boat in still water = $4x = 4 \times 4 = 16$ km/hr.
 Now speed of stream is increased by 50%.
 New speed of stream = $4 \times 150/100$
 New speed of stream = 6 km/hr.
 New upstream speed = Speed in still water – New stream speed
 New upstream speed = $16 - 6$
 New upstream speed = 10 km/hr.
 Required distance upstream = 90 km.
 Time taken = Distance/Speed
 Time taken = $90/10$
 Time taken = 9 hours.

6. Option D

Let the cost price of the bag be Rs. x .
 Marked price is 75% above cost price.
 Marked price = 175% of x .
 Marked price = $175x/100 = 7x/4$.
 Discount allowed = 20%.
 So, selling price = 80% of marked price.

Selling price = $80/100 \times 7x/4$.

Selling price = $7x/5$.

Profit = Selling price - Cost price.

Profit = $7x/5 - x$.

Profit = $7x/5 - 5x/5$.

Profit = $2x/5$.

Given profit = Rs. 800.

So,

$2x/5 = 800$

$2x = 800 \times 5$

$2x = 4000$

$x = 4000/2$

$x = 2000$.

Cost price = Rs. 2000.

Selling price = $7x/5$.

Selling price = $7 \times 2000/5 = \text{Rs. } 2800$.

7. Option A

Let Sandeep's votes be x .

Lokesh gets 25% fewer votes than Sandeep.

So, Lokesh's votes = 75% of x .

Lokesh's votes = $75x/100 = 3x/4$.

Winning margin = Sandeep's votes - Lokesh's votes.

Winning margin = $x - 3x/4$.

Winning margin = $x/4$.

Given winning margin = 900.

So,

$x/4 = 900$

$x = 900 \times 4$

$x = 3600$.

Sandeep's votes = 3600.

Lokesh's votes = $3x/4 = 3 \times 3600/4 = 2700$.

Total votes polled = $3600 + 2700 = 6300$.

8. Option E

Pipes M and N together can fill $1/3$ of the tank in 5 hours.

So, in 10 hours, Pipes M and N will fill double of $1/3$.

Filled part in 10 hours = $\frac{2}{3}$ of the tank.

Pipe S can empty the full tank in 30 hours.

Time taken by Pipe S to empty $\frac{2}{3}$ of the tank = $\frac{2}{3} \times 30$.

Time taken = $60/3 = 20$ hours.

9. Option C

Radius of the circle = 14 cm.

Circumference of circle = $2 \times \pi \times \text{radius}$.

Using $\pi = \frac{22}{7}$:

Circumference = $2 \times \frac{22}{7} \times 14$.

Circumference = 88 cm.

The circumference of the circle is 16 cm less than the perimeter of the square.

So, perimeter of square = $88 + 16 = 104$ cm.

Side of square = Perimeter/4.

Side of square = $104/4 = 26$ cm.

Area of square = side $\times$ side.

Area = $26 \times 26 = 676$ sq. cm.

10. Option B

Ratio of speeds of Train X and Train Y = 5 : 4.

Let the speeds of Train X and Train Y be $5x$ and $4x$ respectively.

Length of Train X = 900 meters.

Length of Train Y = 1260 meters.

When two trains cross each other in opposite directions, total distance covered = sum of their lengths.

Total distance = $900 + 1260 = 2160$ meters.

Relative speed = $5x + 4x = 9x$.

Time taken to cross each other = 24 seconds.

So,

$$9x \times 24 = 2160$$

$$216x = 2160$$

$$x = 2160/216$$

$$x = 10.$$

$$\text{Speed of Train X} = 5x = 5 \times 10 = 50 \text{ m/sec.}$$

11. Option C

In partnership, profit is divided in the ratio of capital $\times$ time.

Investment ratio of M and N = 5 : 8

Let M's investment = $5x$ and N's investment = $8x$.

M's profit share = Rs. 6000

Total profit = Rs. 14000

N's profit share = $14000 - 6000 = \text{Rs. } 8000$

Profit ratio of M and N = $6000 : 8000 = 3 : 4$

M invested for 12 months.

Let N invested for t months.

So, capital-time ratio = $5x \times 12 : 8x \times t$

Capital-time ratio = $60x : 8xt$

This is equal to profit ratio 3 : 4.

So, $60x/8xt = 3/4$

Cancel x .

$60/8t = 3/4$

Cross multiply:

$60 \times 4 = 3 \times 8t$

$240 = 24t$

$t = 240/24 = 10$

12. Option B

Let the cost price of the item be Rs. x .

Marked price is 75% above cost price.

Marked price = 175% of $x = 175x/100 = 7x/4$

Difference between two discounts = $30\% - 18\% = 12\%$

If only 18% discount is given instead of 30%, the selling price increases by 12% of marked price.

Given increase in earning = Rs. 210

So, 12% of marked price = 210

Marked price = $210 \times 100/12$

Marked price = Rs. 1750

Now, marked price = $7x/4$

$7x/4 = 1750$

$x = 1750 \times 4/7$

$x = \text{Rs. } 1000$

13. Option C

Let each person invest Rs. 100.

First person invests at 20% compound interest for 2 years.

Amount after 2 years = $100 \times 120/100 \times 120/100$

Amount = Rs. 144

Compound interest = $144 - 100 = \text{Rs. } 44$

So, first person gets 44% interest.

Second person invests half amount at R% simple interest for 2 years.

Interest on first half = $50 \times R \times 2/100 = R$

Second person invests remaining half amount at 10% simple interest for 2 years.

Interest on second half = $50 \times 10 \times 2/100 = \text{Rs. } 10$

Total interest of second person = $R + 10$

Both interests are equal.

So, $R + 10 = 44$

$R = 44 - 10$

$R = 34$

14. Option B

Pipe P fills half of the tank in 6 minutes.

So, Pipe P fills the full tank in 12 minutes.

Let the efficiency of Pipe P be 1 unit/minute.

Pipe Q is 100% more efficient than Pipe P.

So, efficiency of Pipe Q = 2 units/minute.

Pipe R is 50% less efficient than Pipe Q.

So, efficiency of Pipe R = 50% of 2 = 1 unit/minute.

Efficiency ratio of P, Q and R = 1 : 2 : 1

Together efficiency = $1 + 2 + 1 = 4$ units/minute.

Pipe P alone takes 12 minutes.

When all three work together, time taken = $12/4$

Time taken = 3 minutes

15. Option C

Ratio of quantity of mixture in Vessel P to Vessel Q = 3 : 5

Let quantity in Vessel P = $3x$ liters.

Let quantity in Vessel Q = $5x$ liters.

Milk in Vessel P = 40% of $3x$

Milk in Vessel P = $3x \times 40/100 = 6x/5$

Milk in Vessel Q = 36 liters

Total final mixture = $3x + 5x = 8x$ liters

Final mixture contains 30% milk.

So, total milk in final mixture = 30% of $8x$

Total milk = $8x \times 30/100 = 12x/5$

Now,

Milk from P + Milk from Q = Total milk in final mixture

$6x/5 + 36 = 12x/5$

$36 = 12x/5 - 6x/5$

$36 = 6x/5$

$x = 36 \times 5/6$

$x = 30$

Total final mixture = $8x$

Total final mixture = $8 \times 30 = 240$ liters

16. Option C

Total mixture = 150 liters

Ratio of acid to water = 2:3

Acid = $150 \times 2/5 = 60$ liters

Water = $150 \times 3/5 = 90$ liters

X liters of mixture is removed.

Acid removed = $2X/5$

Water removed = $3X/5$

After removing X liters:

Acid left = $60 - 2X/5$

Water left = $90 - 3X/5$

Now, $(X - 10)$ liters of water is added.

Final water = $90 - 3X/5 + X - 10$

Final water = $80 + 2X/5$

Final ratio of acid to water = 5:9

So:

$(60 - 2X/5) : (80 + 2X/5) = 5 : 9$

$9(60 - 2X/5) = 5(80 + 2X/5)$

$540 - 18X/5 = 400 + 10X/5$

$540 - 400 = 18X/5 + 10X/5$

$140 = 28X/5$

$X = 140 \times 5/28$

$X = 25$

17. Option D

Initial investment ratio of Kunal and Tarun = 4:7

Let Kunal's initial investment = $4x$

Let Tarun's initial investment = $7x$

For the first 6 months:

Kunal's capital-time = $4x \times 6 = 24x$

Tarun's capital-time = $7x \times 6 = 42x$

After 6 months, Kunal increased his initial investment by 50%.

New investment of Kunal = $4x + 50 \times 4x/100$

New investment of Kunal = $4x + 2x = 6x$

After 6 months, Tarun withdrew $3/7$ of his initial investment.

Amount withdrawn by Tarun = $3/7 \times 7x = 3x$

New investment of Tarun = $7x - 3x = 4x$

For the next 6 months:

Kunal's capital-time = $6x \times 6 = 36x$

Tarun's capital-time = $4x \times 6 = 24x$

Total capital-time of Kunal = $24x + 36x = 60x$

Total capital-time of Tarun = $42x + 24x = 66x$

Profit ratio of Kunal to Tarun = $60x : 66x = 10 : 11$

Difference between their profit shares = 11 parts – 10 parts = 1 part

1 part = Rs. 1200

Tarun's profit share = 11 parts = 11×1200

Tarun's profit share = Rs. 13200

18. Option C

Three years ago, average age of Neha, Ritu and Varun = 28 years

Sum of their ages three years ago = $28 \times 3 = 84$ years

Present sum of their ages = $84 + 3 + 3 + 3$

Present sum of their ages = 93 years

Four years hence, sum of ages of Neha and Varun = 68 years

Present sum of ages of Neha and Varun = $68 - 4 - 4$

Present sum of ages of Neha and Varun = 60 years

Present age of Ritu = Present total age – Present age sum of Neha and Varun

Present age of Ritu = $93 - 60$

Present age of Ritu = 33 years

19. Option B

Let Meera's monthly salary be Rs. 100.

Amount spent on rent = 30% of 100 = Rs. 30

Remaining salary = $100 - 30 = \text{Rs. } 70$

Amount spent on groceries = 20% of 70

Amount spent on groceries = $20 \times 70/100 = \text{Rs. } 14$

Saving = $70 - 14 = \text{Rs. } 56$

So, when salary is Rs. 100, saving is Rs. 56.

Given saving = Rs. 11200

Monthly salary = $11200 \times 100/56$

Monthly salary = Rs. 20000

20. Option B

Let downstream speed = $5x \text{ km/hr}$

Let upstream speed = $x \text{ km/hr}$

Time taken to cover $D \text{ km}$ downstream = $D/5x \text{ hours}$

Time taken to cover $D \text{ km}$ upstream = $D/x \text{ hours}$

Total time = 18 hours

So:

$$D/5x + D/x = 18$$

$$D/5x + 5D/5x = 18$$

$$6D/5x = 18$$

$$D/x = 18 \times 5/6$$

$$D/x = 15$$

Speed in still water = $(\text{Downstream speed} + \text{Upstream speed})/2$

$$\text{Speed in still water} = (5x + x)/2$$

$$\text{Speed in still water} = 3x \text{ km/hr}$$

$$\text{Time taken to cover } 3D \text{ km in still water} = 3D/3x$$

$$\text{Time taken} = D/x$$

$$\text{Time taken} = 15 \text{ hours}$$

3. Quadratic Equations

CHECKLIST

In each of the following questions, there are two equations. You have to solve both equations and mark the correct answer.

(a) $x > y$

(b) $x < y$

(c) $x = y$ or the relationship cannot be established

(d) $x \geq y$

(e) $x \leq y$

1.) I. $5x^2 + 3x - 14 = 0$

II. $14y^2 - 55y + 50 = 0$

2.) I. $x^2 - 6\sqrt{10}x + 50 = 0$

II. $y^2 - 6\sqrt{10}y + 80 = 0$

3.) I. $9x^2 - 52x + 35 = 0$

II. $17y^2 + 53y + 6 = 0$

4.) I. $14x^2 - 13x - 1 = 0$

II. $8y^2 - 20y + 12 = 0$

5.) I. $6x^2 + 29x - 42 = 0$

II. $15y^2 + 70y + 40 = 0$

6.) I. $x^2 - 6\sqrt{10}x + 90 = 0$

II. $y^2 - 2\sqrt{10}y + 10 = 0$

7.) I. $12x^2 + 3x - 9 = 0$

II. $7y^2 + 16y + 9 = 0$

8.) I. $11x^2 + 20x - 4 = 0$

II. $4y^2 - 14y + 6 = 0$

9.) I. $7x^2 + 11x + 4 = 0$

II. $5y^2 + 37y + 42 = 0$

10.) I. $x^2 - 2\sqrt{13}x + 13 = 0$

II. $y^2 - 5\sqrt{13}y + 52 = 0$

11.) I. $16x^2 - 12x - 18 = 0$

II. $9y^2 + 3y - 56 = 0$

12.) I. $4x^2 + 30x + 54 = 0$

II. $11y^2 + 43y + 30 = 0$

13.) I. $x^2 - 5\sqrt{11}x + 44 = 0$

II. $y^2 - 2\sqrt{11}y + 11 = 0$

14.) I. $10x^2 + 29x - 21 = 0$

II. $18y^2 - 48y + 24 = 0$

15.) I. $17x^2 + 27x - 14 = 0$

II. $10y^2 + 45y + 50 = 0$

16.) I. $18x^2 - 43x - 5 = 0$

II. $12y^2 + 58y + 18 = 0$

17.) I. $8x^2 + 52x + 80 = 0$

II. $16y^2 + 30y - 25 = 0$

18.) I. $x^2 - 7\sqrt{11}x + 66 = 0$

II. $y^2 - 6\sqrt{11}y + 88 = 0$

19.) I. $13x^2 + 47x + 24 = 0$

II. $6y^2 + 39y + 63 = 0$

20.) I. $15x^2 - 43x + 30 = 0$

II. $3y^2 - 14y + 16 = 0$

Answers:

1. B

2. C

3. A

4. E

5. C

6. A

7. D

8. B

9. A

10. E

11. C

12. E

13. D

14. B

15. D

16. A

17. E

18. C

19. D

20. B

Solutions:

$$(1) x = -2, \frac{7}{5}$$

$$y = \frac{10}{7}, \frac{5}{2}$$

$$(2) x = \sqrt{10}, 5\sqrt{10}$$

$$y = 2\sqrt{10}, 4\sqrt{10}$$

$$(3) x = \frac{7}{9}, 5$$

$$y = -3, -\frac{2}{17}$$

$$(4) x = -\frac{1}{14}, 1$$

$$y = 1, \frac{3}{2}$$

$$(5) x = -6, \frac{7}{6}$$

$$y = -4, -\frac{2}{3}$$

$$(6) x = 3\sqrt{10}, 3\sqrt{10}$$

$$y = \sqrt{10}, \sqrt{10}$$

$$(7) x = -1, \frac{3}{4}$$

$$y = -\frac{9}{7}, -1$$

$$(8) x = -2, \frac{2}{11}$$

$$y = \frac{1}{2}, 3$$

$$(9) x = -1, -\frac{4}{7}$$

$$y = -6, -\frac{7}{5}$$

$$(10) x = \sqrt{13}, \sqrt{13}$$

$$y = \sqrt{13}, 4\sqrt{13}$$

$$(11) x = -\frac{3}{4}, \frac{3}{2}$$

$$y = -\frac{8}{3}, \frac{7}{3}$$

$$(12) x = -\frac{9}{2}, -3$$

$$y = -3, -\frac{10}{11}$$

$$(13) x = \sqrt{11}, 4\sqrt{11}$$

$$y = \sqrt{11}, \sqrt{11}$$

$$(14) x = -\frac{7}{2}, \frac{3}{5}$$

$$y = \frac{2}{3}, 2$$

$$(15) x = -2, \frac{7}{17}$$

$$y = -\frac{5}{2}, -2$$

$$(16) x = -\frac{1}{9}, \frac{5}{2}$$

$$y = -\frac{9}{2}, -\frac{1}{3}$$

$$(17) x = -4, -\frac{5}{2}$$

$$y = -\frac{5}{2}, \frac{5}{8}$$

$$(18) x = \sqrt{11}, 6\sqrt{11}$$

$$y = 2\sqrt{11}, 4\sqrt{11}$$

$$(19) x = -3, -\frac{8}{13}$$

$$y = -\frac{7}{2}, -3$$

$$(20) x = \frac{6}{5}, \frac{5}{3}$$

$$y = 2, \frac{8}{3}$$

4. WRONG NUMBER SERIES

(1) 1432, 1458, 1493, 1544, 1598, 1672

- (a) 1458
- (b) 1598
- (c) 1544
- (d) 1672
- (e) None of these

(2) 142, 299, 911, 3657, 18315, 109793

- (a) 18315
- (b) 911
- (c) 299
- (d) 3657
- (e) None of these

(3) 79654, 39822, 13268, 3320, 654, 100

- (a) 39822
- (b) 100
- (c) 654
- (d) 3320
- (e) None of these

(4) 1820, 1851, 1890, 1903, 1922, 1939

- (a) 1851
- (b) 1890
- (c) 1922
- (d) 1939
- (e) None of these

(5) 210, 236, 458, 718, 1224, 1978

- (a) 210
- (b) 458
- (c) 718
- (d) 1978
- (e) None of these

(6) 850.75, 844.25, 836.5, 821.5, 805.25, 785.75

- (a) 836.5
- (b) 805.25
- (c) 844.25
- (d) 785.75
- (e) None of these

(7) 331, 370, 411, 454, 509, 546

- | | |
|--|---|
| (a) 370 | (e) None of these |
| (b) 454 | |
| (c) 331 | (11) 430, 448, 486, 574, 722, 1040 |
| (d) 509 | (a) 448 |
| (e) None of these | (b) 574 |
| | (c) 722 |
| (8) 515, 1009, 1995, 3975, 7903, 15777 | (d) 1040 |
| (a) 1009 | (e) None of these |
| (b) 3975 | |
| (c) 7903 | (12) 74, 226, 452, 1392, 2800, 8420 |
| (d) 515 | (a) 452 |
| (e) None of these | (b) 226 |
| | (c) 2800 |
| | (d) 8420 |
| (9) 3500, 3482, 3451, 3403, 3344, 3240 | (e) None of these |
| (a) 3482 | |
| (b) 3240 | (13) 2500, 2488, 2457, 2379, 2260, 2040 |
| (c) 3344 | (a) 2040 |
| (d) 3403 | (b) 2488 |
| (e) None of these | (c) 2457 |
| | (d) 2379 |
| (10) 30670, 15360, 5155, 1300, 270, 50 | (e) None of these |
| (a) 15360 | |
| (b) 1300 | (14) 58166, 29079, 9690, 2405, 456, 40 |
| (c) 270 | (a) 456 |
| (d) 50 | (b) 29079 |

(c) 9690

(d) 40

(e) None of these

(15) 125, 150, 210, 336, 614.8, 1209.6

(a) 150

(b) 210

(c) 336

(d) 1209.6

(e) None of these

(16) 720, 742, 771, 809, 868, 920

(a) 742

(b) 771

(c) 868

(d) 920

(e) None of these

(17) 275, 567, 1163, 2329, 4687, 9405

(a) 1163

(b) 567

(c) 4687

(d) 9405

(e) None of these

(18) 255, 288, 323, 360, 389, 440

(a) 288

(b) 323

(c) 440

(d) 389

(e) None of these

(19) 900, 933, 972, 1033, 1080, 1149

(a) 933

(b) 1033

(c) 1080

(d) 1149

(e) None of these

(20) 6400, 3240, 1680, 930, 560, 400

(a) 3240

(b) 1680

(c) 560

(d) 400

(e) None of these

Answers:

(1) c

(2) a

(3) d

(4) b

(5) e

(6) a

(7) d

(8) b

(9) c

(10) e

(11) b

(12) a

(13) d

(14) c

(15) e

(16) c

(17) a

(18) d

(19) b

(20) e

Solutions:

(1) +26, +35, +46, +59, +74; second differences: +9, +11, +13, +15.

(2) $\times 2 + 15$, $\times 3 + 14$, $\times 4 + 13$, $\times 5 + 12$, $\times 6 + 11$.

(3) $\div 2 - 5$, $\div 3 - 6$, $\div 4 - 7$, $\div 5 - 8$, $\div 6 - 9$.

(4) +31, +29, +23, +19, +17.

(5) $210 + 236 + 12 = 458$; $236 + 458 + 24 = 718$; $458 + 718 + 36 = 1212$; $718 + 1212 + 48 = 1978$.

(6) -6.5, -9.75, -13, -16.25, -19.5.

(7) $18^2 + 7$, $19^2 + 9$, $20^2 + 11$, $21^2 + 13$, $22^2 + 15$, $23^2 + 17$.

(8) $\times 2 - 21$, $\times 2 - 23$, $\times 2 - 25$, $\times 2 - 27$, $\times 2 - 29$.

(9) -18, -31, -48, -69, -94; second differences: -13, -17, -21, -25.

(10) $\div 2 + 25$, $\div 3 + 20$, $\div 4 + 15$, $\div 5 + 10$, $\div 6 + 5$.

(11) +18, +38, +78, +158, +318; each next difference = $\times 2 + 2$.

(12) $\times 3 + 4$, $\times 2 + 8$, $\times 3 + 12$, $\times 2 + 16$, $\times 3 + 20$.

(13) $-(2^3 + 4)$, $-(3^3 + 4)$, $-(4^3 + 4)$, $-(5^3 + 4)$, $-(6^3 + 4)$.

(14) $\div 2 - 2^2$, $\div 3 - 3^2$, $\div 4 - 4^2$, $\div 5 - 5^2$, $\div 6 - 6^2$.

(15) $\times 1.2$, $\times 1.4$, $\times 1.6$, $\times 1.8$, $\times 2$.

(16) +22, +29, +38, +49, +62; second differences: +7, +9, +11, +13.

(17) $\times 2 + 17$, $\times 2 + 19$, $\times 2 + 23$, $\times 2 + 29$, $\times 2 + 31$.

(18) 15×17 , 16×18 , 17×19 , 18×20 , 19×21 , 20×22 .

(19) $+(3 \times 11)$, $+(3 \times 13)$, $+(3 \times 17)$, $+(3 \times 19)$, $+(3 \times 23)$.

(20) $\div 2 + 40$, $\div 2 + 60$, $\div 2 + 80$, $\div 2 + 100$, $\div 2 + 120$.

5. MISSING NUMBER SERIES

(1) 58, 67, 85, ?, 164, 233

- (a) 112
- (b) 114
- (c) 116
- (d) 118
- (e) None of these

(2) 7, 19, 48, ?, 242, 509

- (a) 109
- (b) 111
- (c) 113
- (d) 115
- (e) None of these

(3) 1496, 384, ?, 32, 12, 5

- (a) 98
- (b) 100
- (c) 102
- (d) 104
- (e) None of these

(4) 95, 128, 226, ?, 595, 967

- (a) 356
- (b) 360
- (c) 364
- (d) 368
- (e) None of these

(5) 44, 54, 84, ?, 282, 504

- (a) 152
- (b) 154
- (c) 156

(d) 158

(e) None of these

(6) 2750, 2732, 2696, ?, 2480, 2192

- (a) 2618
- (b) 2620
- (c) 2622
- (d) 2624

(e) None of these

(7) 82.5, 86, 91.5, ?, 108.5, 120

- (a) 97
- (b) 98.5
- (c) 99
- (d) 100.5
- (e) None of these

(8) 11, 21, 61, ?, 1201, 7201

- (a) 235
- (b) 237
- (c) 239
- (d) 241
- (e) None of these

(9) 127, 140, 159, ?, 211, 242

- (a) 180
- (b) 182
- (c) 184
- (d) 186
- (e) None of these

(10) 1940, 980, 500, ?, 140, 80

- (a) 250
- (b) 255

- (c) 260
(d) 265
(e) None of these

(11) 3, 14, 39, ?, 155, 258

- (a) 80
(b) 82
(c) 84
(d) 86
(e) None of these

(12) 72, 93, 135, ?, 282, 387

- (a) 194
(b) 196
(c) 198
(d) 200
(e) None of these

(13) 3600, 3582, 3552, ?, 3448, 3370

- (a) 3504
(b) 3506
(c) 3508
(d) 3510
(e) None of these

(14) 35, 66, 123, 230, ?, 834

- (a) 429
(b) 431
(c) 433
(d) 435
(e) None of these

(15) 115, 180, 291, ?, 742, 1189

- (a) 459
(b) 463
(c) 467
(d) 471
(e) None of these

(16) 2400, 2414, 2456, ?, 2960, 4094

- (a) 2582
(b) 2584
(c) 2586
(d) 2588
(e) None of these

(17) 78, 92, 116, ?, 202, 268

- (a) 148
(b) 150
(c) 152
(d) 154
(e) None of these

(18) 2400, 600, 2400, ?, 2400, 400

- (a) 470
(b) 480
(c) 490
(d) 500
(e) None of these

(19) 220, 235, 263, ?, 374, 465

- (a) 304
(b) 306
(c) 308
(d) 310
(e) None of these

(20) 11, 22, 47, ?, 209, 430

- (a) 98
(b) 100
(c) 102
(d) 104
(e) None of these

Answers:

- (1) c
(2) b
(3) d

- (4) b
 (5) a
 (6) d
 (7) c
 (8) d
 (9) b
 (10) c
 (11) c
 (12) c
 (13) c
 (14) d
 (15) b
 (16) a
 (17) c
 (18) b
 (19) c
 (20) b

Solutions:

(1) +9, +18, +31, +48, +69

Second difference: +9, +13, +17, +21

(2) $\times 2 + 5$, $\times 2 + 10$, $\times 2 + 15$, $\times 2 + 20$,
 $\times 2 + 25$

(3) $\div 4 + 10$, $\div 4 + 8$, $\div 4 + 6$, $\div 4 + 4$, $\div 4 + 2$

(4) Sum of previous two numbers +3,
 +6, +9, +12

(5) $+2^3 + 2$, $+3^3 + 3$, $+4^3 + 4$, $+5^3 + 5$,
 $+6^3 + 6$

(6) -18, -36, -72, -144, -288

(7) +3.5, +5.5, +7.5, +9.5, +11.5

(8) $\times 2 - 1$, $\times 3 - 2$, $\times 4 - 3$, $\times 5 - 4$, $\times 6 - 5$

(9) +13, +19, +23, +29, +31

(10) $\div 2 + 10$, $\div 2 + 10$, $\div 2 + 10$, $\div 2 + 10$,
 $\div 2 + 10$

(11) Terms are: $n^3 + n^2 + n$, for $n = 1$,
 2, 3, 4, 5, 6

(12) +21, +42, +63, +84, +105

(13) -2×9 , -3×10 , -4×11 , -5×12 ,
 -6×13

(14) $\times 2 - 2^2$, $\times 2 - 3^2$, $\times 2 - 4^2$, $\times 2 - 5^2$,
 $\times 2 - 6^2$

(15) Sum of previous two numbers
 -4, -8, -12, -16

(16) +14, +42, +126, +378, +1134

Each difference is multiplied by 3.

(17) $+2 \times 7$, $+3 \times 8$, $+4 \times 9$, $+5 \times 10$, $+6 \times 11$

(18) $\div 4$, $\times 4$, $\div 5$, $\times 5$, $\div 6$

(19) $+3 \times 5$, $+4 \times 7$, $+5 \times 9$, $+6 \times 11$, $+7 \times 13$

(20) $\times 2 + 0$, $\times 2 + 3$, $\times 2 + 6$, $\times 2 + 9$, $\times 2 + 12$

6. DATA INTERPRETATION

Set 1: The monthly donation collection target of a relief foundation is Rs. 81 lakhs, and 1% of the total donation target is distributed as a reward fund among various field teams. The table shows the percentage distribution of the reward fund received by different field teams and the number of volunteers in each team. Study the table carefully and answer the questions.

एक राहत फाउंडेशन का मासिक दान संग्रह लक्ष्य 81 लाख रुपये है, और कुल दान लक्ष्य का 1% विभिन्न मैदानी टीमों के बीच पुरस्कार राशि के रूप में वितरित किया जाता है। तालिका विभिन्न मैदानी टीमों को प्राप्त पुरस्कार राशि का प्रतिशत वितरण और प्रत्येक टीम में स्वयंसेवकों की संख्या दर्शाती है। तालिका का ध्यानपूर्वक अध्ययन कीजिए और प्रश्नों के उत्तर दीजिए।

Field Team / मैदानी टीम	Percentage distribution of reward fund / पुरस्कार राशि का प्रतिशत वितरण	Number of volunteers / स्वयंसेवकों की संख्या
Narmada Team / नर्मदा टीम	33 $\frac{1}{3}$ %	54
Yamuna Team / यमुना टीम	16 $\frac{2}{3}$ %	45
Godavari Team / गोदावरी टीम	25%	45
Krishna Team / कृष्णा टीम	8 $\frac{1}{3}$ %	27
Tapti Team / ताप्ती टीम	16 $\frac{2}{3}$ %	90

Q1: What reward amount is given to each volunteer of Narmada Team?

नर्मदा टीम के प्रत्येक स्वयंसेवक को कितनी पुरस्कार राशि दी गई है?

- (A) Rs. 450
- (B) Rs. 400
- (C) Rs. 550
- (D) Rs. 500
- (E) Rs. 300

Q2: If the ratio of reward per volunteer in Godavari Team to reward per volunteer in Yamuna Team is P:Q, then find P + Q.

यदि गोदावरी टीम में प्रति स्वयंसेवक पुरस्कार राशि और यमुना टीम में प्रति स्वयंसेवक

पुरस्कार राशि का अनुपात $P:Q$ है, तो $P + Q$ ज्ञात कीजिए।

- (A) 7
- (B) 5
- (C) 9
- (D) 6
- (E) 8

Q3: Find the average reward amount distributed to Narmada Team, Godavari Team and Tapti Team.

नर्मदा टीम, गोदावरी टीम और ताप्ती टीम को वितरित पुरस्कार राशि की औसत राशि ज्ञात कीजिए।

- (A) Rs. 20250
- (B) Rs. 20050
- (C) Rs. 20350
- (D) Rs. 20150
- (E) Rs. 20450

Q4: Reward per volunteer in Tapti Team is what percentage less than reward per volunteer in Krishna Team?

ताप्ती टीम में प्रति स्वयंसेवक पुरस्कार राशि, कृष्णा टीम में प्रति स्वयंसेवक पुरस्कार राशि से कितने प्रतिशत कम है?

- (A) 30%
- (B) 45%
- (C) 50%
- (D) 35%
- (E) 40%

Q5: What is the maximum reward amount received by a volunteer among all the given teams?

दी गई सभी टीमों में किसी स्वयंसेवक द्वारा प्राप्त अधिकतम पुरस्कार राशि कितनी है?

- (A) Rs. 450
- (B) Rs. 300
- (C) Rs. 500
- (D) Rs. 250
- (E) Rs. 150

Solutions:

CHECKLIST FOR BANK EXAMS 2026 BY AASHISH ARORA

Team / टीम	Reward received / प्राप्त राशि	Volunteers / स्वयंसेवक	Reward per volunteer / प्रति स्वयंसेवक राशि
Narmada / नर्मदा	Rs. 27,000	54	Rs. 500
Yamuna / यमुना	Rs. 13,500	45	Rs. 300
Godavari / गोदावरी	Rs. 20,250	45	Rs. 450
Krishna / कृष्णा	Rs. 6,750	27	Rs. 250
Tapti / ताप्ती	Rs. 13,500	90	Rs. 150

Common values:

Total reward fund = 1% of Rs. 81 lakhs = Rs. 81000

Narmada Team: Reward = $33 \frac{1}{3}\%$ of 81000 = 27000, Volunteers = 54, Per volunteer reward = $27000/54$ = Rs. 500

Yamuna Team: Reward = $16 \frac{2}{3}\%$ of 81000 = 13500, Volunteers = 45, Per volunteer reward = $13500/45$ = Rs. 300

Godavari Team: Reward = 25% of 81000 = 20250, Volunteers = 45, Per volunteer reward = $20250/45$ = Rs. 450

Krishna Team: Reward = $8 \frac{1}{3}\%$ of 81000 = 6750, Volunteers = 27, Per volunteer reward = $6750/27$ = Rs. 250

Tapti Team: Reward = $16 \frac{2}{3}\%$ of 81000 = 13500, Volunteers = 90, Per volunteer reward = $13500/90$ = Rs. 150

1. Option D

Reward received by Narmada Team = Rs. 27000

Number of volunteers in Narmada Team = 54

Reward per volunteer = $27000/54$ = Rs. 500

2. Option B

Reward per volunteer in Godavari Team = Rs. 450

Reward per volunteer in Yamuna Team = Rs. 300

Ratio = 450:300 = 3:2

P + Q = 3 + 2 = 5

3. Option A

Reward distributed to Narmada, Godavari and Tapti = 27000 + 20250 + 13500 = 60750

Average reward = $60750/3$ = Rs. 20250

4. Option E

Reward per volunteer in Tapti Team = Rs. 150

Reward per volunteer in Krishna Team = Rs. 250

Difference = $250 - 150 = 100$

Required percentage = $100/250 \times 100 = 40\%$ less

5. Option C

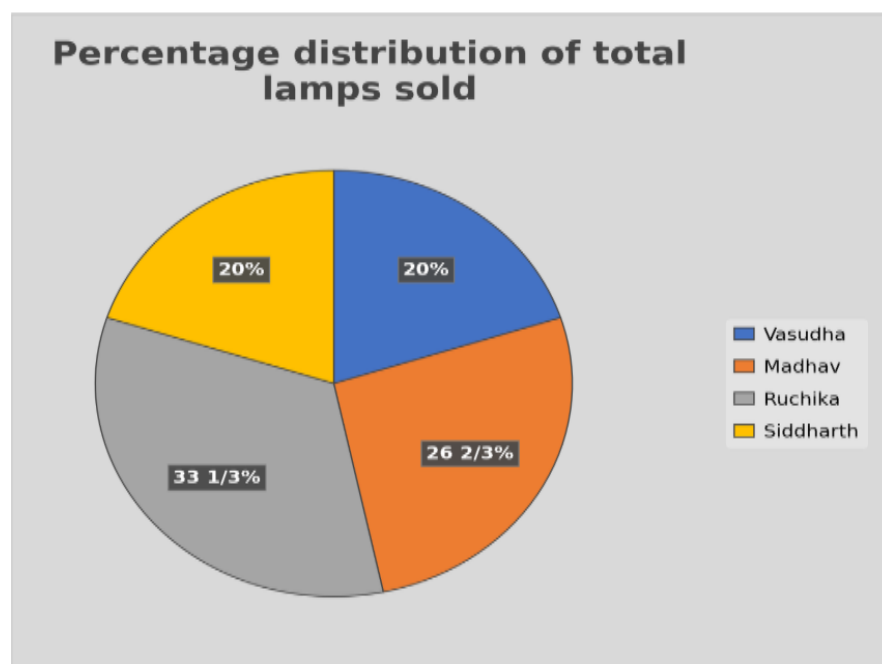
Per volunteer rewards are 500, 300, 450, 250 and 150

Maximum reward per volunteer = Rs. 500

CHECKLIST
BY
AASHISH
ARORA

Set 2: The pie chart shows the percentage or degree distribution of the total number of decorative lamps, table lamps and hanging lamps together, sold by four different showrooms: Vasudha, Madhav, Ruchika and Siddharth. The table-lamp sales of each showroom are 37.5% of the total lamps sold by that showroom. The total number of lamps sold by Siddharth is 720. Read the pie chart carefully and answer the following questions.

पाई चार्ट चार अलग-अलग शोरूम वसुधा, माधव, रुचिका और सिद्धार्थ द्वारा बेचे गए कुल सजावटी लैंप, यानी टेबल लैंप और लटकने वाले लैंप मिलाकर, का प्रतिशत या डिग्री वितरण दर्शाता है। प्रत्येक शोरूम में टेबल लैंप की बिक्री उस शोरूम द्वारा बेचे गए कुल लैंप का 37.5% है। सिद्धार्थ द्वारा बेचे गए कुल लैंप की संख्या 720 है। पाई चार्ट को ध्यान से पढ़िए और निम्नलिखित प्रश्नों के उत्तर दीजिए।



Q1: Hanging lamps sold by a new showroom, Nileshe, are $33 \frac{1}{3}$ % more than the hanging lamps sold by Vasudha, and total lamps sold by Nileshe are 40 less than the average number of total lamps sold by Ruchika and Siddharth. Find the difference between the number of table lamps sold by Madhav and Nileshe.

एक नए शोरूम नीलेश द्वारा बेचे गए लटकने वाले लैंप, वसुधा द्वारा बेचे गए लटकने वाले लैंप से $33 \frac{1}{3}$ % अधिक हैं, और नीलेश द्वारा बेचे गए कुल लैंप, रुचिका और सिद्धार्थ द्वारा बेचे गए कुल लैंप की औसत संख्या से 40 कम हैं। माधव और नीलेश द्वारा बेचे गए टेबल लैंप की संख्या के बीच का अंतर ज्ञात कीजिए।

- (A) 60
(B) 30
(C) 50

- (D) 40
- (E) 70

Q2: Find the ratio between the average number of hanging lamps sold by Vasudha, Ruchika and Siddharth to the number of table lamps sold by Madhav.
वसुधा, रुचिका और सिद्धार्थ द्वारा बेचे गए लटकने वाले लैंप की औसत संख्या का माधव द्वारा बेचे गए टेबल लैंप की संख्या से अनुपात ज्ञात कीजिए।

- (A) 55:36
- (B) 45:32
- (C) 33:22
- (D) 36:55
- (E) 25:18

Q3: Hanging lamps sold by Vasudha and Siddharth together are how much percentage more or less than the total lamps sold by Ruchika?
वसुधा और सिद्धार्थ द्वारा मिलाकर बेचे गए लटकने वाले लैंप, रुचिका द्वारा बेचे गए कुल लैंप से कितने प्रतिशत अधिक या कम हैं?

- (A) 20% more
- (B) 12.5% less
- (C) 25% more
- (D) $33\frac{1}{3}\%$ less
- (E) 25% less

Q4: 37.5% of the total lamps manufactured by Vasudha are sold. If the ratio of hanging lamps to table lamps manufactured by Vasudha is 5:7, then find how much more or less the unsold table lamps are than the unsold hanging lamps of Vasudha.

वसुधा द्वारा बनाए गए कुल लैंप में से 37.5% लैंप बेचे गए हैं। यदि वसुधा द्वारा बनाए गए लटकने वाले लैंप और टेबल लैंप का अनुपात 5:7 है, तो वसुधा के बिना बिके टेबल लैंप, बिना बिके लटकने वाले लैंप से कितने अधिक या कम हैं?

- (A) 450 more
- (B) 500 more
- (C) 480 more
- (D) 520 more
- (E) 400 more

Q5: Floor lamps sold by Madhav are 50% of the total lamps sold by Siddharth. If the ratio of floor lamps to wall lamps sold by Madhav is 9:5, then floor lamps and table lamps sold by Madhav together are what percentage of wall lamps and hanging lamps sold by Madhav together?

माधव द्वारा बेचे गए फ्लोर लैंप, सिद्धार्थ द्वारा बेचे गए कुल लैंप का 50% हैं। यदि माधव द्वारा बेचे गए फ्लोर लैंप और वॉल लैंप का अनुपात 9:5 है, तो माधव द्वारा मिलाकर बेचे गए फ्लोर लैंप और टेबल लैंप, माधव द्वारा मिलाकर बेचे गए वॉल लैंप और लटकने वाले लैंप का कितना प्रतिशत हैं?

- (A) 75%
(B) 84%
(C) 90%
(D) 96%
(E) 80%

Solutions:

Showroom	Total lamps	Table lamps	Hanging lamps	Percentage	Central angle
Vasudha	720	270	450	20%	72°
Madhav	960	360	600	26 2/3%	96°
Ruchika	1200	450	750	33 1/3%	120°
Siddharth	720	270	450	20%	72°
Total	3600	1350	2250	100%	360°

Common Calculation:

Siddharth's share = 20%

20% of total lamps = 720

Total lamps sold = $720 \times 5 = 3600$

Vasudha total lamps = 20% of 3600 = 720

Madhav total lamps = 26 2/3% of 3600 = 960

Ruchika total lamps = 33 1/3% of 3600 = 1200

Siddharth total lamps = 720

Table lamps = 37.5% = 3/8 of total lamps

Vasudha table = $720 \times 3/8 = 270$, hanging = $720 - 270 = 450$

Madhav table = $960 \times 3/8 = 360$, hanging = $960 - 360 = 600$

Ruchika table = $1200 \times 3/8 = 450$, hanging = $1200 - 450 = 750$

Siddharth table = $720 \times \frac{3}{8} = 270$, hanging = $720 - 270 = 450$

1. Option D

Nilesh hanging lamps = $450 \times \frac{4}{3} = 600$

Average total lamps of Ruchika and Siddharth = $(1200 + 720)/2 = 960$

Nilesh total lamps = $960 - 40 = 920$

Nilesh table lamps = $920 - 600 = 320$

Madhav table lamps = 360

Required difference = $360 - 320 = 40$

2. Option A

Hanging lamps of Vasudha, Ruchika and Siddharth = 450, 750 and 450

Average = $(450 + 750 + 450)/3 = 550$

Madhav table lamps = 360

Required ratio = $550:360 = 55:36$

3. Option E

Hanging lamps of Vasudha and Siddharth = $450 + 450 = 900$

Total lamps sold by Ruchika = 1200

Difference = $1200 - 900 = 300$

Required percentage = $300/1200 \times 100 = 25\%$ less

4. Option B

Vasudha sold lamps = 720

37.5% of manufactured lamps = 720

Total lamps manufactured by Vasudha = $720 \times \frac{8}{3} = 1920$

Manufactured hanging lamps = $1920 \times \frac{5}{12} = 800$

Manufactured table lamps = $1920 \times \frac{7}{12} = 1120$

Unsold hanging lamps = $800 - 450 = 350$

Unsold table lamps = $1120 - 270 = 850$

Required difference = $850 - 350 = 500$ more

5. Option C

Floor lamps sold by Madhav = 50% of 720 = 360

Floor lamps : wall lamps = 9:5

Wall lamps sold by Madhav = $360 \times \frac{5}{9} = 200$

Madhav table lamps = 360

Madhav hanging lamps = 600

Required percentage = $(360 + 360)/(200 + 600) \times 100 = 720/800 \times 100 = 90\%$

CHECKLIST

BY

AASHISH

ARORA

Set 3: The table shows the ratio of wind speed to the speed of a drone in still air and the speed of each drone while flying with the wind. Read the data carefully and answer the following questions.

तालिका हवा की गति और शांत हवा में ड्रोन की गति के अनुपात तथा प्रत्येक ड्रोन की हवा के साथ उड़ने की गति को दर्शाती है। डेटा को ध्यानपूर्वक पढ़िए और निम्नलिखित प्रश्नों के उत्तर दीजिए।

Note: Speed with wind = Speed of drone in still air + Wind speed.

Speed against wind = Speed of drone in still air - Wind speed.

नोट: हवा के साथ गति = शांत हवा में ड्रोन की गति + हवा की गति।

हवा के विरुद्ध गति = शांत हवा में ड्रोन की गति - हवा की गति।

Drone (ड्रोन)	Wind Speed : Drone Speed in Still Air (हवा की गति : शांत हवा में ड्रोन की गति)	Speed with Wind (km/h) हवा के साथ गति (किमी/घंटा)
Aadi (आदि)	10 : 40	50
Bhanu (भानु)	10 : 35	45
Charu (चारु)	15 : 45	60
Dev (देव)	10 : 50	60
Esha (ईशा)	10 : 25	35

Q1: Drone Aadi takes 12 hours to cover equal distances against the wind and with the wind. Find the total distance covered by Aadi.

ड्रोन आदि हवा के विरुद्ध और हवा के साथ समान दूरी तय करने में 12 घंटे लेता है। आदि द्वारा तय की गई कुल दूरी ज्ञात कीजिए।

- (A) 420 km
- (B) 480 km
- (C) 450 km
- (D) 500 km
- (E) 540 km

Q2: The distance covered against the wind by Drone Bhanu and Drone Charu is the same. If the ratio of time taken by Charu to Bhanu is $p : q$, then find the value of $(p + q)$.

ड्रोन भानु और ड्रोन चारु द्वारा हवा के विरुद्ध तय की गई दूरी समान है। यदि चारु और भानु द्वारा लिए गए समय का अनुपात $p : q$ है, तो $(p + q)$ का मान ज्ञात कीजिए।

- (A) 11

- (B) 9
- (C) 13
- (D) 15
- (E) 17

Q3: Drone Bhanu travels 225 km with the wind and 225 km against the wind. Drone Charu travels 360 km with the wind and 360 km against the wind. Find the difference between the total time taken by Bhanu and Charu.

ड्रोन भानु हवा के साथ 225 किमी और हवा के विरुद्ध 225 किमी उड़ता है। ड्रोन चारु हवा के साथ 360 किमी और हवा के विरुद्ध 360 किमी उड़ता है। भानु और चारु द्वारा लिए गए कुल समय के बीच अंतर ज्ञात कीजिए।

- (A) 8 hours
- (B) 6 hours
- (C) 2 hours
- (D) 4 hours
- (E) 10 hours

Q4: The speed of Drone Manav in still air is 20 km/h more than that of Drone Bhanu. The wind speed for Manav is half of the wind speed for Aadi. Find the time taken by Manav to cover 360 km with the wind.

शांत हवा में ड्रोन मानव की गति, ड्रोन भानु की गति से 20 किमी/घंटा अधिक है। मानव के लिए हवा की गति, आदि के लिए हवा की गति की आधी है। मानव द्वारा हवा के साथ 360 किमी दूरी तय करने में लिया गया समय ज्ञात कीजिए।

- (A) 4 hours
- (B) 6 hours
- (C) 8 hours
- (D) 5 hours
- (E) 9 hours

Q5: If the speed of Drone Charu in still air increases by 33.33% while the wind speed remains unchanged, then find the distance covered by Charu in 4 hours while flying with the wind.

यदि शांत हवा में ड्रोन चारु की गति 33.33% बढ़ जाती है और हवा की गति अपरिवर्तित रहती है, तो हवा के साथ उड़ते हुए चारु द्वारा 4 घंटे में तय की गई दूरी ज्ञात कीजिए।

- (A) 260 km
- (B) 280 km
- (C) 320 km

(D) 340 km

(E) 300 km

Solutions:

Drone (ड्रोन)	Wind Speed हवा की गति	Still-air Speed शांत हवा में गति	Speed with Wind हवा के साथ गति	Speed against Wind हवा के विरुद्ध गति
Aadi (आदि)	10	40	50	30
Bhanu (भानु)	10	35	45	25
Charu (चारु)	15	45	60	30
Dev (देव)	10	50	60	40
Esha (ईशा)	10	25	35	15

Useful values:

Aadi: Wind speed = 10, Still-air speed = 40, Speed with wind = 50, Speed against wind = 30

Bhanu: Wind speed = 10, Still-air speed = 35, Speed with wind = 45, Speed against wind = 25

Charu: Wind speed = 15, Still-air speed = 45, Speed with wind = 60, Speed against wind = 30

Dev: Wind speed = 10, Still-air speed = 50, Speed with wind = 60, Speed against wind = 40

Esha: Wind speed = 10, Still-air speed = 25, Speed with wind = 35, Speed against wind = 15

1. Option CLet the one-side distance be d km.Time against wind = $d/30$ Time with wind = $d/50$

$$d/30 + d/50 = 12$$

$$(5d + 3d) / 150 = 12$$

$$8d/150 = 12$$

$$d = 225$$

Total distance = $225 + 225 = 450$ km

2. Option A

Speed against wind of Bhanu = 25 km/h

Speed against wind of Charu = 30 km/h

For the same distance, time ratio is inverse of speed ratio.

Time taken by Charu : Time taken by Bhanu = $1/30 : 1/25$

Ratio = $25 : 30 = 5 : 6$

$p + q = 5 + 6 = 11$

3. Option D

Bhanu's total time = $225/45 + 225/25$

Bhanu's total time = $5 + 9 = 14$ hours

Charu's total time = $360/60 + 360/30$

Charu's total time = $6 + 12 = 18$ hours

Required difference = $18 - 14 = 4$ hours

4. Option B

Bhanu's still-air speed = 35 km/h

Manav's still-air speed = $35 + 20 = 55$ km/h

Aadi's wind speed = 10 km/h

Manav's wind speed = $10/2 = 5$ km/h

Manav's speed with wind = $55 + 5 = 60$ km/h

Required time = $360/60 = 6$ hours

5. Option E

Charu's still-air speed = 45 km/h

$33.33\% = 1/3$

New still-air speed of Charu = $45 \times 4/3 = 60$ km/h

Wind speed remains = 15 km/h

New speed with wind = $60 + 15 = 75$ km/h

Distance covered in 4 hours = $75 \times 4 = 300$ km

Set 4: The data given below shows the expenses of three diagnostic centres, Tarun Diagnostic Centre, Kritika Diagnostic Centre and Naveen Diagnostic Centre. For Tarun Diagnostic Centre, the total salary expense given to 112 employees is ₹15.68 lakh and the total miscellaneous expense is ₹47.04 lakh. For Kritika Diagnostic Centre, the average salary of employees is ₹700 less than that of Tarun Diagnostic Centre. The total expense of Kritika Diagnostic Centre and Naveen Diagnostic Centre is the same, and each is 25% less than the total expense of Tarun Diagnostic Centre. The total salary expense of Kritika Diagnostic Centre is ₹2.94 lakh more than that of Tarun Diagnostic Centre. For Naveen Diagnostic Centre, the number of employees is 25% less than that of Tarun Diagnostic Centre and the total miscellaneous expense is ₹31.36 lakh.

Note:

Average salary = Salary expense ÷ Total employees

Total expense = Salary expense + Miscellaneous expense

नीचे दिया गया डेटा तीन डायग्नोस्टिक सेंटर, तरुण डायग्नोस्टिक सेंटर, कृतिका डायग्नोस्टिक सेंटर और नवीन डायग्नोस्टिक सेंटर के खर्चों से संबंधित है। तरुण डायग्नोस्टिक सेंटर में 112 कर्मचारियों को दिया गया कुल वेतन खर्च ₹15.68 लाख है और कुल विविध खर्च ₹47.04 लाख है। कृतिका डायग्नोस्टिक सेंटर में कर्मचारियों का औसत वेतन, तरुण डायग्नोस्टिक सेंटर के औसत वेतन से ₹700 कम है। कृतिका डायग्नोस्टिक सेंटर और नवीन डायग्नोस्टिक सेंटर का कुल खर्च समान है, और प्रत्येक का कुल खर्च तरुण डायग्नोस्टिक सेंटर के कुल खर्च से 25% कम है। कृतिका डायग्नोस्टिक सेंटर का कुल वेतन खर्च, तरुण डायग्नोस्टिक सेंटर के कुल वेतन खर्च से ₹2.94 लाख अधिक है। नवीन डायग्नोस्टिक सेंटर में कर्मचारियों की संख्या, तरुण डायग्नोस्टिक सेंटर की तुलना में 25% कम है और कुल विविध खर्च ₹31.36 लाख है।

नोट:

औसत वेतन = वेतन खर्च ÷ कुल कर्मचारी

कुल खर्च = वेतन खर्च + विविध खर्च

Q1: Find the average of the average salary of employees of Tarun Diagnostic Centre, Kritika Diagnostic Centre and Naveen Diagnostic Centre in rupees.

Approximate value is required.

तरुण डायग्नोस्टिक सेंटर, कृतिका डायग्नोस्टिक सेंटर और नवीन डायग्नोस्टिक सेंटर के कर्मचारियों के औसत वेतन का औसत रुपये में ज्ञात कीजिए। अनुमानित मान ज्ञात कीजिए।

(A) ₹15180

(B) ₹15440

(C) ₹15322

(D) ₹15610

(E) ₹15200

Q2: Find the difference between the total number of employees in Kritika Diagnostic Centre and Naveen Diagnostic Centre.

कृतिका डायग्नोस्टिक सेंटर और नवीन डायग्नोस्टिक सेंटर में कर्मचारियों की कुल संख्या के बीच अंतर ज्ञात कीजिए।

(A) 56

(B) 42

(C) 49

(D) 63

(E) 70

Q3: The total expense of Tarun Diagnostic Centre is what percentage more than the miscellaneous expense of Naveen Diagnostic Centre?

तरुण डायग्नोस्टिक सेंटर का कुल खर्च, नवीन डायग्नोस्टिक सेंटर के विविध खर्च से कितने प्रतिशत अधिक है?

(A) 80%

(B) 90%

(C) 120%

(D) 110%

(E) 100%

Q4: If the average salary in Naveen Diagnostic Centre is increased by 25% and the total expense remains the same, then the new miscellaneous expense is how much more or less than the miscellaneous expense of Kritika Diagnostic Centre?

यदि नवीन डायग्नोस्टिक सेंटर में औसत वेतन 25% बढ़ा दिया जाए और कुल खर्च समान रहे, तो नया विविध खर्च, कृतिका डायग्नोस्टिक सेंटर के विविध खर्च से कितना अधिक या कम होगा?

(A) ₹84000 less

(B) ₹98000 less

(C) ₹112000 more

(D) ₹70000 less

(E) ₹126000 more

Q5: The total number of employees in Devansh Diagnostic Centre is equal to the average number of employees in Tarun Diagnostic Centre, Kritika Diagnostic Centre and Naveen Diagnostic Centre. The average salary of employees in

Devansh Diagnostic Centre is ₹17500. If the ratio of miscellaneous expense to salary expense is 5 : 2, then find the total expense of Devansh Diagnostic Centre in ₹000.

देवांश डायग्नोस्टिक सेंटर में कर्मचारियों की कुल संख्या, तरुण डायग्नोस्टिक सेंटर, कृतिका डायग्नोस्टिक सेंटर और नवीन डायग्नोस्टिक सेंटर के कर्मचारियों की औसत संख्या के बराबर है। देवांश डायग्नोस्टिक सेंटर में कर्मचारियों का औसत वेतन ₹17500 है। यदि विविध खर्च और वेतन खर्च का अनुपात 5 : 2 है, तो देवांश डायग्नोस्टिक सेंटर का कुल खर्च ₹000 में ज्ञात कीजिए।

- (A) 6510
(B) 7000
(C) 6720
(D) 6860
(E) 7140

Solutions:

Centre	Employees	Salary Expense (₹000)	Misc. Expense (₹000)	Total Expense (₹000)
Tarun Diagnostic Centre	112	1568	4704	6272
Kritika Diagnostic Centre	140	1862	2842	4704
Naveen Diagnostic Centre	84	1568	3136	4704
Total	336	4998	10682	15680

1. Option C

Average salary of Tarun = $1568 \times 1000 \div 112$

= ₹14000

Average salary of Kritika = $1862 \times 1000 \div 140$

= ₹13300

Average salary of Naveen = $1568 \times 1000 \div 84$

= ₹18666.67

Required average = $(14000 + 13300 + 18666.67) \div 3$

= ₹15322.22

Approximate value = ₹15322

2. Option A

Employees in Kritika Diagnostic Centre = 140

Employees in Naveen Diagnostic Centre = 84
 Required difference = $140 - 84$
 $= 56$

3. Option E

Total expense of Tarun Diagnostic Centre = 6272
 Miscellaneous expense of Naveen Diagnostic Centre = 3136
 Required percentage = $(6272 - 3136) \div 3136 \times 100$
 $= 100\%$

4. Option B

Salary expense of Naveen Diagnostic Centre = 1568
 New salary expense = 125% of 1568
 $= 1960$
 Total expense remains = 4704
 New miscellaneous expense = $4704 - 1960$
 $= 2744$
 Miscellaneous expense of Kritika Diagnostic Centre = 2842
 Required difference = $2842 - 2744$
 $= 98$ thousand
 $= ₹98000$ less

5. Option D

Employees in Devansh Diagnostic Centre = $(112 + 140 + 84) \div 3$
 $= 112$
 Salary expense = 112×17500
 $= ₹1960000$
 Salary expense = 1960 in ₹000
 Miscellaneous : Salary = 5 : 2
 2 parts = 1960
 1 part = 980
 Miscellaneous expense = 5×980
 $= 4900$
 Total expense = $1960 + 4900$
 $= 6860$ in ₹000